

Shared Mental Models and Boundary Management Activities in New Product Development: A Perspective on Multiteam Systems

Yang Lei , Xiaosong Wu , and James Jiang 

Abstract—Multiteam systems (MTSs) have been increasingly adopted as an organizational form to manage large-scale projects. However, new product development (NPD) in the MTSs structure entails challenges with respect to achieving efficiency due to interteam coordination challenges. Shared mental models (SMMs) within a single team have been found to facilitate intrateam coordination; however, further exploration is needed as to whether SMMs within MTSs can improve interteam coordination among component teams and, if so, what mechanisms underlie this. The present article examines the theoretical underpinning of the impacts of SMMs among component team leaders on MTSs' NPD efficiency, and further considers the impact of external conditions of requirement uncertainty, the most common external situation during NPD. Using survey data from 110 information technology development MTSs and drawing on boundary management perspective, the results reveal that SMMs among component team leaders positively affect NPD efficiency by managing the MTSs boundary, i.e., through boundary entrainment and boundary buffering activities at the system level. In addition, we find that requirement uncertainty plays a vital role in determining the nature of this relationship. Taken together, our findings highlight the importance of the role of SMMs among component team leaders in facilitating NPD efficiency in MTSs settings.

Index Terms—Boundary management perspective, multiteam systems (MTSs), new product development (NPD) efficiency, requirement uncertainty, shared mental models (SMMs) theory.

Manuscript received 18 May 2021; revised 2 September 2021, 10 November 2021, and 4 January 2022; accepted 7 January 2022. Date of publication 24 February 2022; date of current version 29 November 2023. This work was supported in part by the Key Program of NSFC-FRQSC Joint Project under Grant 72061127002, in part by the NSFC project under Grant 72032006, Grant 71731009, and Grant 92146005, in part by the Ministry of Science and Technology of the Taiwan under Grant 109-2410-H-002-074-MY3, in part by the National Center for Applied Mathematics Shenzhen under Grant 518055, and in part by the Shenzhen Key Research Base in Arts and Social Sciences (Intelligent Management and Innovation Research Center-IMIDRC, SusTech, Shenzhen). Review of this manuscript was arranged by Department Editor L. Santiago. (Yang Lei and Xiaosong Wu are co-first authors.) (Corresponding author: James Jiang.)

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This article has supplementary material provided by the authors and color versions of one or more figures available at <https://doi.org/10.1109/TEM.2022.3147905>.

Digital Object Identifier 10.1109/TEM.2022.3147905

I. INTRODUCTION

IN today's business environment, with large-scale projects and the complexities of work demand increasing to such an extent that many are beyond the management capacity of a single team, organizations are turning toward multiteam systems (MTSs) to accomplish multifaceted tasks [1]. An MTS is "two or more teams that interface directly and interdependently in response to environmental contingencies toward the accomplishment of collective goals" [2]. For example, in new product development (NPD), various component teams are called upon to work interdependently with other component teams [3], and they may involve at different stages of the whole NPD process as Team A's work may rely on what Team B has done. One of the most significant challenges for NPD, as for other MTSs' projects, is project efficiency, as cost overrun and schedule overrun are common [4] due to issues such as resource limitations, distinct component team objectives, and increasing demand for coordination [5]. Unlike in standalone teams, where intrateam collaboration is required, MTSs require more interteam coordination to combine the efforts of various component teams responsible for operational task execution [6], which can be particularly challenging given that there exist significant differences in component teams' routines and thought-worlds [7]. Ideally, what influences a successful team would also influence a successful MTS. However, this may not be the reality since the processes that make teams successful may not be the same processes that facilitate successful MTSs [8].

Indeed, MTSs have been considered as a unique level of analysis that prior traditional team-level theory is unable to fully explain [9], [10]. A few studies have devoted initial efforts to understand the determinants of interteam coordination in MTSs, such as intrapersonal functional diversity, nonleader level boundary spanning (interpersonal interactions among members), and the overall MTSs leadership [1], [6], [11]–[13]. While these studies broaden our knowledge on the interteam coordination in MTSs, others point out the component team leaders, acting as the bridge connecting component team with the MTS, would be an important facilitator for the coordination within MTSs [14]. However, there lacks empirical studies along this line to enrich our knowledge. According to the coordination literature that highlights the crucial role of a shared cognition of individuals in facilitating coordination [15], we posit that a shared cognition among each component team leader of an MTS should be helpful for enabling interteam coordination.

The shared mental models (SMMs) theory has been adopted to explain intrateam coordination mechanisms that makes teamwork efficiently [16]. It proposes that SMMs, which refer to the knowledge shared by members of a team when cooperating on team tasks [17], are strong predictors of team efficiency because of the contribution they make to internal team coordination [18]. However, whether SMMs in the MTSs have the same effect remains unclear. In the MTSs that are structured according to two levels, the component-team level and the system level [8], [10], the SMMs are distinct in their meanings. At the component-team level, SMMs entail that team members in a same team organize their knowledge of specific team tasks, equipment, roles, goals, and abilities in a similar fashion, which allow them to anticipate others' actions in the same component. At the system level, SMMs refer to that each component team leader, who is the representative of each component team' project-based understanding, has a clear understanding of other teams' progresses and the information about the knowledge repertoire that other component teams hold, through which they could coordinate behaviors accordingly. The differences between SMMs at the two levels include the fact that subjects at the system level utilize different project management processes, have their own team values, and pursue their own proximal goals [19], which is not the case for team members within a single team [1]. Thus, although the effects of SMMs at the component-team level on intrateam coordination could be understood by the implications of prior single team studies [20], existing literature has shed little light on the influence of SMMs at the system level, i.e., SMMs among component team leaders, on the efficiency of MTSs' project considering the challenging problems of interteam coordination. In addition, prior literature has found that the influence of SMMs on team efficiency and team performance is through the enhanced team process [18]. However, whether such findings fit the MTSs context have not been empirically tested. Therefore, it is crucial to investigate whether SMMs among component team leaders have a significant impact on interterm coordination as well as NPD efficiency—and, if so, how.

To answer the question, we first look deeply at the mechanisms of SMMs on NPD efficiency by adopting a boundary management perspective. The boundary management perspective, which emphasizes that a team should facilitate both entrainment activities within its borders and buffering activities outside its borders, serves as a solid theoretical foundation for us to understand how various kinds of information are processed in team process [21]. Specifically, entrainment is an inward-oriented information processing activity, referring to adjusting the pace or cycle of an activity to match or synchronize with that of another activity within a team [22], while buffering is an outward-oriented information processing activity that a team undertakes to filter external interference, such as change in the external environment [23]. Similarly, for a project that requires between-team coordination and encounters specific requirements from external customers, the MTSs as a whole should not only facilitate entrainment activities among component teams but also buffer change in the external environment at the system level. In addition, the MTS-based complex NPD often face with requirement uncertainty, which has been found as one important contingent factor to influence the effectiveness of SMMs [24].

However, we still have little empirical examination on the effect of requirement uncertainty on SMMs at the system level.

To sum up, this article aims to explain the effect of SMMs among MTSs leader team on NPD efficiency from the boundary management perspective under requirement uncertainty. Because a high degree of information-processing complexity is inherent in MTSs work, both within and outside the MTSs, a leader team that is comprised of component team leaders is often formed, and is overseen by the MTSs leader. The leader team serves to handle aspects that require coordination among the component teams or exchanges with the external environment beyond the MTSs [11], wherein the MTSs leader is responsible for the overall progress of the MTSs project and for communicating with customers, while the leaders of component teams are responsible for the progress of their own tasks.

Our article makes several contributions to theory. First, we contribute to the NPD literature by empirically investigating the antecedents of NPD efficiency from the MTSs perspective. While a few conceptual studies have proposed its implications for NPD [25], [26], empirical evidence of the MTSs perspective in the project management disciplines is still underexplored [27]. This article thus adds to the emerging MTSs literature and provides a foundation for future researchers to focus on investigating NPD efficiency, or any other project efficiency, from an MTSs perspective. Thus, practitioners, who are currently focusing on improving single-team efficiency in large-scaled projects, are suggested to manage such projects from such a perspective, that is to improve the overall efficiency of the MTSs. Second, this article expands SMMs theory into the MTSs setting and provides a preliminary theoretical framework for establishing a specific MTSs–SMMs theory by investigating SMMs at the system level (e.g., SMMs among component team leaders) and the underlying mechanisms. Specifically, our article provides insight into the mediating role of MTSs process, namely both the entrainment and buffering mechanisms at the system level, confirming their distinct effects in promising MTSs-based NPD efficiency. Therefore, rather than focusing on each component team's proximal goals distinctly, component team leaders are suggested also to have a clear understanding of other teams' progress and specializations to keep interterm coordination. In addition, component teams within an MTSs should keep in pace with each other and also protect the whole project from external influence. Third, this article shows how requirement uncertainty influences the relationship between SMMs and NPD in an MTS setting, thereby emphasizing the importance of considering requirement uncertainty in order to determining the effects of SMMs on boundary management. Thereby, besides managing within MTSs activities, the whole MTSs should assess the level of requirement uncertainty. Especially when requirement uncertainty is high, SMMs show a greater power in determining NPD efficiency.

II. THEORETICAL FOUNDATION

A. MTSs Perspective on New Product Development

Prior articles have identified several features that distinguish MTSs from traditional standalone teams [2], [8]: first, MTSs are composed of at least two component teams, each of which

has its own proximal goals. Second, component teams share a common goal—that is, completing the overall MTSs' project. Third, the structure or configuration of an MTS is determined by its goals, which are organized into a hierarchy with proximal team goals at the lowest level and distal MTSs' goals at the highest level. Fourth, component teams have input, process, or outcome interdependence with at least one other team in the MTSs.

Conceptual studies have emphasized the importance of researching MTSs as a unique level of analysis [19], and have recognized the crucial role of interterm coordination among various component teams in MTSs. Subsequently, scholars devote their efforts in researching the antecedents of interterm coordination in the MTSs context. First, leadership of the MTSs has been identified as a crucial determinant of interterm coordination within an MTS. For example, DeChurch and Marks [11] find that MTSs functional leadership is positively related with interteam coordination, which further influence MTSs performance. Luciano *et al.* [28] proposed that strategic leadership system is positively related to MTSs performance. In addition, besides examining the influence of MTSs leaders, a personal perspective is adopted to investigate how component team members influence the interteam coordination. For example, De Vries *et al.* [6] studied how intrapersonal functional diversity of component team members influence MTSs performance through interteam coordination, of which they documented a positive result. Ziegert *et al.* [12] and Dietrich *et al.* [13] documented that interpersonal interactions among MTSs members would benefit the entire MTSs performance. Although these articles have broadened our knowledge on the interteam coordination in MTSs, it is still lack of knowledge on how component team leaders play their roles. This article aims to fill this gap by introducing the role of SMMs in MTSs leader team, involving MTSs leaders and component team leaders.

NPD lies beyond the managing capacity of a single team; instead, the NPD process has been considered by organizational scholars to entail an MTS structure that usually involves multiple skilled component teams [29], including manufacturing, design, sales, etc. For example, NPD in IT firms involves a hardware team, a software team, a customer support team, an accounting and finance team, and so on [30]. The question of how to improve NPD efficiency—that is, to increase speed and reduce cost—has long been subject to attention from scholars in the fields of project management. For example, Jaber *et al.* [31] investigated the crucial role of efficient collaborative decisions within teams in improving NPD projects, while Choi and Phan [32] explored how a firm's exploration and exploitation activities influence NPD performance. In addition, in a meta-analysis, Chen *et al.* [33] indicated that factors such as team leadership and internal integration play crucial roles in improving NPD efficiency.

However, as noted above, although extant research has broadened our understanding on NPD, most articles have been conducted at the single team or organizational level. We posit that ignoring MTSs nature of the NPD may hinder both academics and practitioners to have a complete picture. We believe that it is necessary to investigate the influencing factors of NPD efficiency from an MTS perspective in which multiple component

teams work together. Within the fields of project management, the adoption of the MTSs perspective is in its infancy [27]. With a few conceptual papers on investigation of NPD from the MTSs perspective [25], limited empirical studies have used it to investigate NPD in terms of component team leaders' behaviors and interteam collaborations on large-scale projects.

B. Shared Mental Models Theory

Mental models, which are defined as the “mechanism whereby humans are able to generate descriptions of system purpose and form, explanations of system functioning and observed system states” [34], help people to describe, explain, and predict events in their environment. Consequently, as individuals are often organized into teams to accomplish tasks, SMMs have been proposed to explain team functioning. SMMs are defined as the knowledge shared by members of a team when cooperating on team tasks [9]. The idea is that team performance improves if team members have a shared understanding of the task that is to be performed and of the teamwork involved.

Deeply, prior articles point out that SMMs are a complex cognitive concept which includes two major domains—task-related SMMs and team-related SMMs—where teams develop both teamwork and taskwork track behavior [35]. Specifically, task-related SMMs refer to the shared understanding of the team's specific work/task, such as their predictions, judgment criteria, and perceptions of progress regarding the task [18]. Team-related SMMs entail shared understanding on team structure including team members' constitution, skills, and team expertise distribution, which is beneficial for any members to coordinate and cooperate with other teammates to accomplish the work [18].

Prior articles have established that the impact of SMMs on team processes and performance is robust at the single-team level [36]. In the MTSs structure, the SMMs will be more important but complex, reflected at two aspects, respectively, levels and components. First, SMMs could be achieved into two levels. At the component-team level, SMMs entail team members organizing their knowledge of specific team tasks, equipment, roles, goals, and abilities, which allow them to anticipate others' actions in the same component team. At the system level, SMMs refer to that each component team leader has a clear understanding of other teams' tasks, actions, and then coordinate behaviors accordingly. We do believe that the SMMs literature provides insight to the effects of SMMs at the component-team level, but is not suitable for explaining the effect of SMMs at the system level. However, limited empirical studies have been conducted to explore the impacts of SMMs at the system level on MTSs performance, despite several conceptual and case studies [37], [38].

Therefore, this article seeks to broaden our knowledge by examining the influence of SMMs at the system level, i.e., SMMs among component team leaders. In addition, following prior articles [35], we conceptualize SMMs at the system level in the MTSs structure as a second-order construct, including both task-related and team-related SMMs. This extension is reasonable because component team leaders in the context of NPD

need to share component team development progress and report the problems with those of interdependent component teams as well as MTSs leaders for coordination and cooperation [11], which means that even though component teams may commit to their local component team work [6], they still need to make sure their task and activities compatible to others. Therefore, we would like to define the system-level task-related SMMs as each component team leader has a clear shared understanding of interdependent teams' work progress, procedures, and policies. In addition, component teams facing development problems need to scout expertise around the MTSs, which make it necessary for component team leaders to recognize the expertise of other teams and locate them when required. Thus, system-level team-related SMMs are defined as each component team leader has a clear shared understanding of other teams' expertise and the whole MTSs expertise distribution.

C. Boundary Management

1) *Information-Processing Approach of Boundary Management*: This article adopted the information-processing approach to understand the boundary management in the MTSs context. While the literature has pointed out three different approaches to manage boundary, information-processing boundary, interpretive boundary, and political boundary [39], information-processing boundary has been studied as the dominant view used to describe managing boundaries in product development [40]. Such approach emphasizes the importance of boundary management issues, both within-team boundary activities, known as entrainment, and outside-team boundary activities, known as buffering, for predicting team performance. Specifically, entrainment is an inward-oriented activity, which is conceptualized as the work pace and cycle alignment of one given component teams to the MTS-level "pace givers" (e.g., project-level milestones and associated reviews) and also adjustment with other component teams [41], thereby coordinating cycles and pace within a team; buffering is an outward-oriented activity, which focuses on the information processing of external environment, including absorbing useful knowledge and protecting the team from external interference [23]. For example, in order to achieve a shared goal, a team must orchestrate its interrelated actions and responses [22] and maintain its boundaries [42].

The MTSs boundary management faces both internal information processing and external information processing, as well. Specifically, the MTSs need to not only synchronize cycles and pace between component teams, but also protect the MTSs project from external interference. Indeed, two levels of information-processing boundaries exist: the system boundary, which delineates the whole MTSs from its external environment; and the component team boundary, which separates a component team from others within the MTSs. Prior article on standalone team studies has investigated the influence of boundary entrainment [43] and boundary buffering [23] activities on performance and efficiency. Such literature may help us to understand the boundary management problem at the component team level. However, boundary management at the system level in an MTS structure has received limited attention. In addition, considering

rapid changes that occur in the external environment, such as requirement uncertainty from customers, it is crucial for MTSs to set appropriate boundaries. Therefore, our article seeks to fill this gap and broaden our knowledge along this line.

2) *Requirements Uncertainty*: Team boundary activities may also be challenged by the level of uncertainty, i.e., uncertainty of information that the team faces in the external environment. In the NPD context, this is also a significant issue, as it is common for customers to change their requirements about the product's function, size, or shape, or even to decide to suspend the project at any point in the development lifecycle in response to a dynamic environment [44]. From the information-processing perspective, requirements uncertainty refers to the asymmetric information between necessary information that need to identify customers' requirements and the amount of information possessed by the MTSs [45]. Further, three important dimensions of requirement uncertainty has been identified: requirement instability, which refers to the extent of changes in customers' requirements along the project life cycle; requirement diversity, which relates to distinct requirements among customers; and requirement analyzability, which refers to the extent to which customers' needs can be converted to a set of requirement specifications [46]. However, while the moderating effect of uncertainty has been investigated in traditional team context [23], [24], [47], we still have little empirical examination on the effect of requirement uncertainty on SMMs at the system level. Considering that high level of requirement uncertainty may lead to more information to be processed, a higher level of SMMs may alleviate such external interference through encouraging more boundary management activities. Therefore, we argue that requirement uncertainty may moderate the relationships between SMMs and boundary management activities at the system level.

In summary, we would expect that information-processing approach of boundary management provides a solid theoretical foundation to investigate how the SMMs among component team leaders achieve NPD efficiency through enhancing boundary management activities under level of requirement uncertainty.

III. HYPOTHESES DEVELOPMENT

Drawing on SMMs theory, we focus on SMMs among component team leaders. Adopting information-processing approach of boundary management perspective, we propose two different mechanisms, namely entrainment and buffering at the system level, to explain how SMMs among component team leaders influence NPD efficiency under levels of requirement uncertainty. The detailed hypothesis development is shown in Fig. 1.

A. SMMs and NPD Efficiency

We posit that the SMMs in the MTS leader team involving leaders of component teams and MTS leader would positively influence the overall outcome of an MTS—that is, NPD efficiency. In an MTS setting, many of the tasks are interdependent. When the leaders of component team have a high level of SMMs, they could grasp more information on other teams' progress, and have the information of how and who to effectively handover

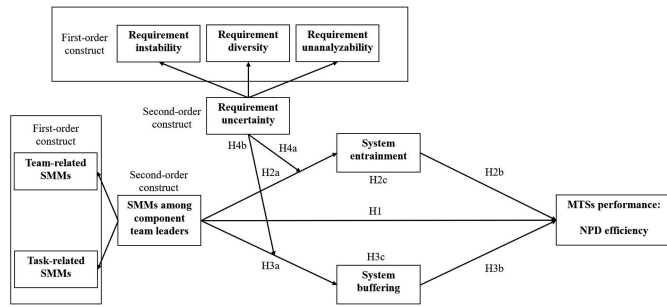


Fig. 1. Research model.

the deliverables. As a result, they will more readily work on interdependent tasks, which improve the outcomes of the MTSs. For example, for an MTS software product development project, the software development team needs to rely on the results of the design team, while the testing team relies on the work of the software development team. If the SMMs among these component teams are high, such that each team is aware of the information of the others' IT module deliverables and related handover procedure, they would be more readily to follow-up, thus improving the overall development efficiency. In addition, if the structure of knowledge, i.e., each component team is aware of the other teams' expertise, is shared, when one team encounters difficulties in solving problems, they could get help immediately by finding the right helper, which would also facilitate the develop efficiency. Therefore, we put forward the following hypothesis:

H1: SMMs in an MTS leader team are positively associated with NPD efficiency.

B. SMMs, System Entrainment, and NPD Efficiency

As we illustrated above, in a complex NPD project, each component team is responsible for a portion of subtasks that are interrelated according to complexity and professional requirements [48]. This task complementary creates interdependency among the component teams in terms of business knowledge, development practices, testing data, system development documentation, quality metrics, and deliverables handover [49]. A high degree of SMMs among leaders of component teams means interdependent component teams for different IT module development could follow standard and agreed procedures and policies to handover the IT module deliverables, which can improve temporal coordination (entrainment) as the shared knowledge and information could reduce the communication cost and improve the efficiency of inward-oriented information processing activities [50]. For example, on the one hand, entrainment involves fitting with the rhythm of other component teams [22]. This fit is more likely to be achieved if the leaders of each component team are "on the same page" regarding the specifics of their associated IT modules delivered by other component teams and knowing other component teams' expertise [51]. Specifically, only after component team leaders share a common understanding of the interdependent task, it is possible for them to adapt their behaviors to fit with each other. In addition,

the shared understanding of MTSs knowledge (e.g., the whole MTSs schedule) distribution and component teams expertise (e.g., find the professional members for solving problems as soon as possible) can also help to direct each component team to align with each other's schedule and talent/resource usage for reducing any interteam conflict and time waste [52], which have been both found to have a negative influence on entrainment activities [53]. This is particularly the case in the context of NPD, which can involve multiple component teams with a common goal but different values. Thus, we propose that a high degree of SMMs among the leaders of component teams in general improves entrainment at the system level in an MTS.

We also posit that entrainment activities at the system level could improve NPD efficiency. First, with more entrainment activities, each component team will have greater capability to back other teams up, which has been shown to have a positive influence on overall project efficiency [54]. In addition, entrainment activities can ease the learning process within an MTS, which benefits performance [55]. Component teams can learn from each other more easily by using entrainment activities to generate new ideas. Consequently, by recombining existing knowledge with new ideas, an MTS for NPD can bring new levels of innovation to organization products and services. For example, Burt [56] suggested that those who are adept at innovating through a recombination of knowledge have a "vision advantage" over others in the organization. He also suggested that this vision advantage comes from bridging structural gaps in an organization's communication network. Therefore, entrainment activities (e.g., collaboration) can further improve NPD efficiency. On the basis of these considerations, we posit that SMMs among leaders of component teams will improve NPD efficiency because they strengthen entrainment activities at the system level. In summary:

H2a: SMMs in an MTS leader team are positively associated with entrainment activities at the system level.

H2b: Entrainment activities are positively associated with NPD efficiency.

H2c: Entrainment activities mediate the positive effect of SMMs in an MTS leader team on NPD efficiency.

C. SMMs, System Buffering, and NPD Efficiency

In addition to entrainment activities, and in response to the increased complexity of work tasks and changing environmental conditions, an MTS must increasingly conduct buffering activities across its boundaries and actively manage relationships with external stakeholders [42]. We posit that SMMs in MTS leader team could help the MTSs to accurately determine whether external inputs are beneficial or necessary for the MTSs and then more cohesively respond to external inputs. In a complex IT NPD project, the MTSs may encounter unwanted influences and interruptions from the external environment. For example, MTSs may be influenced by ambassador activities from top levels of the organization [57] and unreasonable IT requirements from the customers. Thus, the MTS leader team would protect the MTS project from these interferences and requirements in order to maintain the pace of the MTSs as a whole [58]. When the leaders of component teams share a high degree of share common goals

regarding the overall MTSs, they are clearer on what the MTSs and each component team needs or do not need from outside the MTSs. Thus, the SMMs could enhance the MTSs' ability to process external information [59], conducting buffering activities to protect the MTSs from unwanted influences and interruptions. With a low degree of SMMs among component leaders, the MTSs will be vulnerable to external influences with distinct component team goals and each component team leaders may not conduct buffering for avoiding prevent beneficial and important information flowing into the MTSs. Thus, we hypothesize that SMMs among leaders of component teams are positively associated with buffering activities at the system level.

We believe that buffering activities at the system level further improve NPD efficiency, as studies have found that buffering activities enhance team performance [60], [61]. In an MTS context, especially for NPD, buffering activities not only protect component teams from external interference so that these teams can move quickly, but also increase the quality and success of the product. For example, when faced with unreasonable changes or unexpected requirements from customers, buffering can strengthen the boundaries against such external interference and create a protected internal atmosphere for the component teams, thereby accelerating NPD. Therefore, we hypothesize that buffering at the system level is positively associated with NPD efficiency. On the basis of the foregoing discussion, we also propose that SMMs among leaders of component teams in the MTSs will improve NPD efficiency because they enhance buffering activities at the system level.

H3a: SMMs in an MTS leader team are positively associated with buffering activities at the system level.

H3b: Buffering activities are positively associated with NPD efficiency.

H3c: Buffering activities mediate the positive effect of SMMs in an MTS leader team on NPD efficiency.

D. Moderating Effects of Requirement Uncertainty

Requirement uncertainty will improve the effect of SMMs on the entrainment and buffering by creating more additional coordination and buffering task from the perspective of information processing, which enhance the role of SMMs. In this article, requirement uncertainty includes the requirement instability, diversity, and analyzability [46]. When the requirement instability is high, it means customer will frequently change their IT requirements [47]. In this situation, the MTSs could either follow or refuse the changed requirements. If the MTSs follow the changed requirements, then each component team needs to adapt its own team works/tasks and schedules to the new global IT requirements and further coordinate with each other [62], which requires more additional entrainment to attain this adaption and coordination. As above, the SMMs provide established and agreed coordination procedures for component team and enable more entrainment activities. Therefore, the higher requirement instability, the more additional entrainment activities will be enabled by established and agreed coordination procedures via SMMs. On the other hand, if the MTSs refuse the IT requirements changes (e.g., due to high cost and risk for this change), especially for unreasonable requirements, the MTSs

need to conduct more additional buffering to protect the MTSs and components teams to respond to changes. As above, the SMMs provide established and agreed guidance on determining how and who to buffer. Thus, the higher requirement instability, the more buffering enabled by the same SMMs.

When the requirement diversity is high, it means inconsistent and even conflicting requirements to be proposed by a group of users [63]. In this case, the MTSs need to work on solving any requirement conflicts by additional entrainment or buffering these conflicting requirements, asking the group of users to discuss by themselves for working out a more consistent IT requirement. Similarly, the additional entrainment or buffering demands created by the requirement diversity will enhance the role of SMMs on realizing more entrainment and buffering activities. When the requirement unanalyzability is high, it means the requirements proposed by customers are ambiguous and hard to be translated into detailed specifications [64]. Then, the MTSs in whole need are to fully utilize the expertise of different component teams and integrate them to translate part of the IT requirements. For those too ambiguous requirements, necessary buffering is also required to conduct for requiring the customers for further clarity. Studies have shown that uncertainty increases the demand for a high level of information processing [65]. To cope with such demand, the MTSs should enhance its information-processing capabilities by means of cultivating SMMs among component team leaders. Thus, alignment between information-processing demands and capabilities produces better outcomes [66]. Similarly, Mathieu *et al.* [18] argued that SMMs are more important for coordination in an MTS when workload, time pressure, or external requirement uncertainty are high. In the face of changing customer requirements frequently, workload will increase and a low degree of SMMs may present a barrier to component teams predicting each other's information and resource requirements. Given the above considerations, we propose the final pair of hypotheses:

H4a: Requirement uncertainty positively moderates the relationship between SMMs in an MTS leader team and entrainment activities at the system level.

H4b: Requirement uncertainty positively moderates the relationship between SMMs in an MTS leader team and buffering activities at the system level.

IV. METHODOLOGY

A. Samples and Data Collection

To test the above hypotheses, we collected data using a matched-pair field survey. The samples for the survey comprised IT development MTSs within IT firms located in China. Given that complex IT systems or software development tasks are often carried out in the form of MTSs and suffer from requirement changes, we believe that focusing on NPD in the context of IT development is appropriate for testing our hypotheses. With the support of a known IT association, the survey targeted 500 IT companies in China chosen at random. Of the companies invited in our research project, 320 agreed. A questionnaire package, including a cover letter with instructions, the survey, and postage-paid return envelopes, was sent to each firm's

coordinator. To avoid potential correlation errors among multiple MTSs in the same company, we asked each company to base its responses on only one recently completed IT development project that took the form of an MTS, regardless of the project's success or failure.

To minimize the risk of common method variance, the matched-pair survey was conducted using a multisource research design targeting two types of respondents: MTS leaders and component team leaders. The MTS leaders, who assemble a leader team consisting of all component team leaders, are responsible for NPD efficiency, team-related SMMs, and buffering at the system level, because they are in charge of the overall IT development MTSs. The component team leaders account for significant portions of the specific development task by coordinating with other team leaders, and are more familiar with the level of between-team entrainment achieved and the extent of task-related SMMs. Furthermore, their greater familiarity with any requirement changes in IT development puts them in a good position to answer questions about the level of requirement uncertainty faced by the MTSs. Accordingly, two types of separate and confidential questionnaire were sent to each type of respondent; these are the asked specific MTS-related questions and questions on the respondents' demographic characteristics.

Matching the different informants' separate responses with unique identification numbers yielded 132 matched responses. Of those, 22 surveys were discarded because of incomplete responses and outliers (two cases with Cook distance values above 0.6), and the final sample consisted of the remaining 110 observations. Thus, the usable response rate was 22%, which is comparable to other studies using a matched-pair survey with senior executives [67]. In order to examine the degree of potential bias, we used late respondents as surrogates for a sample of nonrespondents and compared the differences in the means for early and late respondents. The results showed no significant differences between early respondents and late respondents in relation to the key constructs. The demographics of sampled firms, MTS, MTS leaders, and component team leaders are summarized in the table of Online Appendix A.

B. Measurement Development

The contextual operationalization of the research constructs serves as a starting point for developing the measurement scale. Existing validated measures of requirement uncertainty and NPD efficiency were adopted, and requirement uncertainty as a second-order construct was measured via its three dimensions—requirement instability, requirement diversity, and requirement unanalyzability [46]. For the two types of boundary activities, entrainment and buffering at the system level, existing measures were adapted to suit our context of IT development in an MTS. Given the lack of a suitable scale for measuring SMMs at the system level, including team-related and task-related components, we developed new measures (see online Appendix B for the items). Our development of SMMs began with collecting original items from numerous prior articles. We first identified the frame of SMMs measurement to summarize two strategies: One is to directly ask for the extent of the mental models to

be shared. Another is to assume that the mental model will be widely shared, such that less effort during coordination should be expected. Both strategies require identifying the knowledge content of the mental model that should be focused on. In this article, we adopted the first strategy for our team-related mental model and considered the expertise distribution and who grasps what expertise as the knowledge content of the mental model, based on the work of [68]. On the other hand, we adopted the second strategy for our task-related mental model measurement, and considered the knowledge of different dimensions of IT product complexity that requires collaboration as the content of task-related SMMs, based on [69]. In combining and adapting these previous related measures, we made sure that our measures incorporated both task- and team-related SMMs. We originally developed 15 items, which we reduced to six through a three-round expert review process (interviews with 2 IT managers, 10 sorters in two-step Q-sorting, and a pilot study). All items were worded such that they referred to the MTS, not the individual or single team, because the critical measure is the perceived degree to which mental models are shared among the leader team. In our model, SMMs are a second-order factor with two task- and team-related SMMs as formative first-order factors. The organizational and MTSs characteristics of organization size, MTSs size, and MTSs duration were included as important controls for potential alternative explanations for the hypothesized relationships. All measurement items were scored on a five-point Likert scale (1 = very weak or strongly disagree, 5 = very strong or strongly agree).

To ensure the validity of these adapted and self-developed measures in the context of an IT development MTSs, and to capture unambiguously the domain of each construct, all items were refined through a process of expert review. The survey was translated into Chinese by one of the researchers, and the translation was checked by a second researcher fluent in both English and Chinese. Two experienced managers responsible for an IT development MTS reviewed the Chinese version of the survey for clarity, and helped to modify any problematic items and improve content validity. To improve the face validity and convergent validity, two-step Q-sorting was conducted to ensure that all facets of the relevant construct were included and that the items for each construct were convergent [70]. Two separate rounds of sorting (with five sorters in each round) classified the previously generated items on the basis of their similarity to definitions and descriptions of the underlying construct categories. In the first round, the accuracy rate for item-to-construct definition classifications was around 80%. After minor corrections based on agreement between the sorters' classifications, the modified items were classified by a different group of five sorters in the second round, during which 90% of the items were successfully classified into the intended constructs. These improvements indicate that face validity and convergent validity were established. Confident that the items were reliable, we proceeded to pilot test the survey on a convenience sample of 20 MTS leaders who had experience of managing a customized complex IT development MTSs. The items were minimally modified based on feedback from this pilot study. All the final measurement items are listed in Online Appendix B.

TABLE I
CORRELATION MATRIX AMONG DIRECTLY OBSERVED CONSTRUCTS

Constructs	ESMM	ASSM	RI	RD	RU	Entra	Buff	Effi
Team-related SMMs	0.86							
Task-related SMMs	0.21*	0.91						
Requirement instability	-0.05	0.04	0.88					
Requirement Diversity	0.08	0.08	0.76*	0.86				
Requirement unanalyzability	-0.03	0.17	0.23*	0.21*	0.82			
System entrainment	0.19	0.43*	-0.001	-0.05	0.33*	0.84		
System buffering	0.41	0.01	0.12	0.04	0.09	0.15	0.85	
NPD efficiency	0.57*	0.42*	0.08	0.13	0.23*	0.62*	0.44*	0.90

Note: $N = 110$. The diagonal elements in this correlation matrix are the square root of the average variance extracted; * $p < 0.05$.

V. DATA ANALYSIS AND RESULTS

Ordinary least squares (OLS) analysis was used to test the hypotheses. Before the OLS analysis, we used the latent factor score approach to compute a single indicator for our two composite constructs, requirement uncertainty, and SMMs. We first tested the validity of the measurements of the directly observed first-order constructs of SMMs and the first-order constructs of requirement uncertainty to assess whether their factor structures were valid for the proposed conceptual model. Next, we calculated the latent factor scores of each first-order construct of requirement uncertainty and SMMs. Finally, we calculated the single indicator of requirement uncertainty and SMMs using their respective first-order construct factor scores and repeating the latent factor score approach.

A. Measurement Model

We began by assessing the reliability and validity of our measurement instrument, testing 1) item reliability, 2) construct reliability, and 3) measurement validity. First, all factor loadings were above the acceptable threshold (0.70). Second, the Cronbach's alpha values for the constructs consisting of multiple indicators were higher than the recommended threshold (0.70). These results confirmed that all of the formative multi-indicator constructs had satisfactory reliability (see Online Appendix C for details of the reliability analysis). Third, the validity of all measurements was tested using an explorative factor analysis with principal component analysis (see Online Appendix C for the cross-loadings). All the items loaded more highly on the intended construct than on any other construct, providing evidence of discriminant and convergent validity among the multi-indicator constructs. Table I shows the correlations among constructs.

We tested discriminant validity using the average variance extracted (AVE). Discriminant validity is inferred when the square root of each construct's AVE is higher than the correlation of the construct to other latent variables [71]. All of the diagonal values in Table I, which are the square roots of each construct's AVE, were much higher than the construct's correlations with other constructs. These results provide additional evidence of

discriminant validity among the multi-indicator constructs. LISREL 8.7 software was then used to run a confirmatory factor analysis (CFA) for the measurement model. The CFA model fit indices of the measurement model were within accepted guidelines, confirming the goodness of fit of our measurement model (chi-square = 281.12; $df = 247$; incremental fit index, 0.97; non-normed fit index, 0.97; comparative fit index, 0.97). In addition, the root mean square error of approximation (RMSEA) was 0.036; an RMSEA of less than 0.08 is considered practical evidence of good model fit [72].

On the basis of these validity results for the directly observed research constructs, the SMMs construct and requirement uncertainty are, respectively, formed using the latent factor scores, their first-order constructs as their new formative indicators, and further used for further hypothesis tests.

B. Hypothesis Testing

1) *Testing of Main and Mediating Effects:* The hypothesized relationship of the main and mediating effects in the research model was tested using Model 4 of the PROCESS macro of SPSS 20.0. All control variables were included in the model. After 5000 times of bootstrapping, the results for the main effects and mediating effects are computed to be reported in Table II. (As we found no significant effect for the control variables, these results are not included.)

First, there was a significant effect of SMMs on NPD efficiency without involving entrainment and buffering at the system level ($\beta = 0.392$, $p < 0.05$), providing support for our H1. Second, there was a significant effect of SMMs on both system entrainment ($\beta = 0.388$, $p < 0.05$) and system buffering ($\beta = 0.269$, $p < 0.05$). Thus, H1a and H2a are supported. Third, there was a significant effect of both system entrainment ($\beta = 0.422$, $p < 0.05$) and system buffering ($\beta = 0.275$, $p < 0.05$) on NPD efficiency, which provides support for H1b and H2b. Finally, in the presence of both system entrainment and system buffering, the effect of SMMs on NPD efficiency was reduced, although it was still significant. Taken together, these results indicate that both system entrainment ($\beta = 0.164$, $p < 0.05$) and system buffering ($\beta = 0.074$, $p < 0.05$) act as two important mediators

TABLE II
RESULTS OF TESTING FOR MAIN AND MEDIATING EFFECTS

	Model pathways	Estimated	95% CI		Results
			Lower	Upper	
System entrainment	SMMs→Efficiency (without mediator)	0.392	0.260	0.524	S
	SMMs→ System Entrainment	0.388	0.210	0.566	S
	System Entrainment→Efficiency	0.422	0.293	0.551	S
	SMMs→Efficiency (with mediator)	0.164	0.130	0.371	S
System buffering	SMMs→ Efficiency (without mediator)	0.392	0.260	0.524	S
	SMMs→ System Buffering	0.269	0.082	0.456	S
	System Buffering→ Efficiency	0.275	0.152	0.398	S
	SMMs→ Efficiency (with mediator)	0.074	0.024	0.152	S

Notes: S = significant.

TABLE III
HIERARCHICAL REGRESSION MODEL

	System entrainment			System buffering		
	Model 1	Model2	Model 3	Model 1	Model2	Model 3
Block 1: Control variables						
Organization Size	-0.03(0.10)	0.01(0.08)	-0.03(0.10)	0.03(0.10)	0.05(0.10)	0.02(0.10)
MTSs size	-0.01(0.16)	0.001(0.15)	-0.05(0.15)	0.10(0.16)	0.10(0.15)	0.05(0.15)
MTSs duration	0.15(0.11)	0.12(0.01)	0.11(0.10)	0.06(0.11)	0.03(0.11)	0.02(0.11)
Block 2: Main effect						
SMMs		0.36*(0.09)	0.29*(0.10)		0.31*(0.09)	0.23*(0.10)
Block 3: Moderating effect						
Requirement uncertainty			0.08(0.09)			0.08(0.10)
SMMs × requirement uncertainty			0.23* (0.10)			0.23*(0.11)
ΔR^2		0.130	0.039		0.092	0.040
R^2	0.022	0.152	0.191	0.017	0.109	0.149
F	0.805	4.704*	4.064*	0.620	3.210*	3.010*

Note: Number of observations (n) is 110. All path coefficients are standardized. The value in each parenthesis is the standard error. * $p < 0.05$.

between SMMs and NPD efficiency. Thus, H1c and H2c are supported.

2) *Test of Moderating Effects*: The hypothesized relationship for the moderating effects in the research model was tested by means of hierarchical regression analysis with OLS using SPSS 20.0. We first incorporated the control variables, independent variables, moderators, and interaction term into our research model to check the variance of the two mediators as dependent variables. Then, we assessed the significance level of the model by calculating the increased explained variance and F hierarchical value [73]. Last, three models were formed for each mediator as the dependent variable, as shown in Table III.

In Model 1, the three control variables (organization size, MTSs size, and MTSs duration) have an insignificant effect on both system entrainment and system buffering. Model 2 explains

the significant effect of SMMs on both system entrainment and system buffering in the complex IT development MTS. SMMs thus positively influence both system entrainment and system buffering. In Model 3, requirement uncertainty has significant and positive moderating effects on both the relationship between SMMs and system entrainment ($\beta = 0.23, p < 0.05$), and the relationship between SMMs and system buffering ($\beta = 0.23, p < 0.05$). These significant moderating effects are further verified by the significance level of the F hierarchical value. Thus, H4a and H4b are supported.

VI. DISCUSSION

The results confirm all of our hypotheses: first, SMMs among leaders of component teams in an MTS are positively associated

with NPD efficiency, through the two system-level boundary management activities in MTSs structure. This finding reveals that when the leaders of each component team share information about their own team's actions on the task, as well as those of other component teams within the MTSs, the MTSs will conduct more boundary management activities at the system level—that is, entrainment and buffering activities—which in turn improve the efficiency of the whole MTSs.

Second, our results emphasize the importance of the two boundary management activities at the system level. From the interterm process perspective, entrainment is an inward-oriented information processing activity that keep each component team in pace, and buffering is an outward-oriented information processing activity that keep the MTSs away from external interference. In addition, it is worth noting that the mediating effects of both boundary entrainment and boundary buffering are only partial. This indicates that besides these two system-level boundary management activities, there may exist other variables that link SMMs among team leaders to MTSs efficiency. For example, a higher level of SMMs among component team leaders may improve the MTSs commitment of all component teams, which is beneficial for MTS performance [74]. In addition, the interteam process within an MTS—that is, boundary management activities at the component-team level—may be influenced by the SMMs among team leaders and further impact NPD efficiency.

The finding that requirement uncertainty positively moderates the relationships between SMMs and entrainment and buffering is in accordance with those of previous qualitative studies [18]. The positive effects of SMMs on boundary entrainment and buffering activities exist only when requirement uncertainty is high. As mentioned above, when customer demand is more uncertain, the MTSs need not only to process more information but also to cope with extra work and time pressure. Thus, SMMs among the leaders of component teams have a greater impact in situations of high requirement uncertainty, while in situations of low requirement uncertainty, SMMs among component team leaders are less important but still significantly contribute to both boundary management activities.

A. Theoretical Implications

The findings of our article have several theoretical contributions. First, our article contributes to the NPD literature by broadening understanding of the antecedents of NPD efficiency from a single-team perspective to an MTS perspective. As the MTS perspective has been characterized as a unique level of analysis in organizations [7], [75], a few conceptual studies have proposed its implications for NPD [25], [26]. However, a lack of relevant empirically MTSs research has hindered a comprehensive understanding of certain topics. Given that the MTSs perspective is relatively novel in project management areas [27], this article adds to the emerging MTSs literature and encourages future researchers to focus on improving NPD efficiency from an MTS perspective.

Second, our article has implications for the body of knowledge on SMMs in terms of adapting SMMs theory to an MTS

perspective. SMMs theory has been widely adopted to predict NPD performance at the team level [20]. Even with growing acknowledgment in the literature, as well as in practice, of the importance of MTSs in NPD projects, there have been limited empirical attempts to examine SMMs in relation to NPD efficiency at the system level. Our article adapts SMMs theory to the MTS level of analysis and differentiates SMMs at the two levels of an MTS structure; therein, we find evidence that SMMs among leaders of component teams can improve NPD efficiency. Given the unique characteristics of MTSs, our article represents an initial empirical effort to extend the application of SMMs theory, and provides a point of reference for its future development. This has wider implications for the application of SMMs theory, especially in the operations management discipline. In addition, by adopting an information processing approach on boundary management, our research provides a more comprehensive picture of the mechanism by which SMMs at the system level influence NPD efficiency. While prior articles have documented the direct effects of SMMs on team performance [76], less effort has been dedicated to investigating the underlying mechanisms [77]. By extending the boundary management perspective to the system level of the MTSs context, our findings document that both entrainment and buffering activities at the system level play mediating roles between SMMs among the leaders of component teams and NPD efficiency. Consequently, we claim that the boundary management perspective provides an important and useful theoretical framework for supplementing SMMs theory and enriching behavior operations research.

Finally, this article contributes to the literature on uncertainty in the fields of project management. We incorporate and test how requirement uncertainty influences the relationship between SMMs and NPD at the system level. Although the literature has emphasized that requirement uncertainty is an important contingency factor in teamwork and performance [24], very few studies have empirically tested its contingent role in the relationship between SMMs and NPD efficiency. Therefore, this article advances knowledge of how the relationship between SMMs and NPD efficiency changes with the degree of external uncertainty. Building on theoretical discussions in previous research regarding how SMMs operation relates to task demand, our article provides strong empirical evidence that SMMs among the leaders of component teams in MTSs become more important when customer demand is highly unpredictable and less effective when requirement uncertainty is low.

B. Practical Implications

In addition to its theoretical implications, this article has a number of practical implications for managers and team leaders of MTSs. First, the positive relationship between SMMs among leaders of component teams in an MTS and NPD efficiency emphasizes the importance of SMMs at the system level. While most project leaders are currently focusing on improving single-team efficiency, it is recommended that large-scale IT project managers should provide leaders of component teams with ample opportunities to establish SMMs. SMMs cannot be constructed automatically; it takes time to start working together,

to gather teamwork experience, and to learn from mistakes. For example, Xiang *et al.* [76] found that awareness and management of other people's emotions are crucial antecedents of SMMs. Therefore, large-scale project managers are suggested to encourage each component team leader to become involved in more extra-role and off-job activities, which are beneficial for establishing SMMs.

Second, in a large-scale project involving MTSs, such as NPD, boundary management activities are crucial for success. It is recommended that large-scale project managers encourage both coordination within MTSs and adaptation to the outside environment. Within the MTSs boundary, entrainment activities at the system level should be emphasized as this may help to improve the efficiency of the overall MTSs. For example, project managers should focus on the cycle, pace, dominant rhythms, and time orientation of each component team to enact successful entrainment [22]. In addition, a balance of boundary buffering activities should be encouraged [23] because these protect the team from external influence and allow useful knowledge to be absorbed. Therefore, practices to implement buffering activities must be established, and should include metrics to determine their efficacy.

Third, the results of our article have practical implications for large-scale project managers with respect to requirement uncertainty from customers. We find that when requirement uncertainty is high, the effects of SMMs among component team leaders are more significant. Therefore, although requirement uncertainty may not be explicit, the MTS leader should assess it, to the best of their ability, at the beginning of the MTSs project. In addition, because of the dynamic characteristic of requirement uncertainty, the MTS leader is suggested to continue to assess requirement uncertainty at different stages of the MTSs project.

C. Limitations and Future Research

This article is subject to several limitations that should be considered in future research. First, the results are based on data from a sample of Chinese firms. Data collection from a single culture limits the generalization of research findings to other cultures, and future articles should seek to take account of cultural differences. Second, our data collection methodology limits further investigation of the dynamic nature of large-scale projects, which project management scholars regard as explaining the different boundary activities required and the changes to their relative importance during the project life cycle [42]. For example, when the interdependence of each component team becomes more reciprocal, the uncertainty between component teams increases. Therefore, a longitudinal research design would provide stronger evidence with respect to whether the relationships identified in our article vary over the stages of a large-scale project. Third, we confirmed the significance role of SMMs among team leaders of component teams in leading to NPD efficiency; however, we did not explore the different functioning of SMMs among component team leaders, such as the accuracy and similarity of SMMs [78], [79]. As prior articles have revealed that SMMs are crucial within standalone teams,

future research could further broaden our knowledge of SMMs within the MTSs structure by specifically focusing on that angle.

Last, this article investigated the mediating effects of boundary management activities at the system level in an MTS structure. Such boundary management activities—entrainment and buffering activities—are crucial in enhancing NPD efficiency. However, our article is subject to two limitations in this regard: On the one hand, we did not investigate the boundary management activities at the component team level, of which future research could have a deeper investigation; on the other hand, prior articles have documented that there exist different knowledge boundaries in the NPD process [39]. For example, Carlile [80] identified three types of knowledge boundary: an information-processing boundary that emphasizes the transfer of knowledge; an interpretive boundary that emphasizes the translation of knowledge; and a political boundary that emphasizes the transformation of knowledge. This article only focuses on the first type of knowledge boundary between cross-functioning teams (component teams), and emphasizes information processing. Therefore, we encourage future articles to further augment this line of research by considering the other two types of knowledge boundary.

VII. CONCLUSION

In summary, this article investigates the effects of SMMs among component team leaders on NPD efficiency through enhanced entrainment and buffering activities at the system level of MTSs under levels of requirement uncertainty, based on information-processing lens of boundary management. It finds that SMMs among component team leaders are positively related with the two boundary management activities at the system level, which in turn improve NPD efficiency. In addition, the effects of SMMs are more pronounced in situations of high requirement uncertainty. These findings not only enrich our knowledge along the project management and NPD literature, but also offer greater clarity regarding how NPD could be achieved efficiency for practitioners.

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